Saam Haghighat-Grami

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Roswell, GA
Interests – Autonomous, Mobile, Collaborative Robotics

Graduating December 2026
LinkedIn – saam-haghighat-grami
GitHub – Saam-Grami
Portfolio – saam-grami.github.io
Active Security Clearance

Education

- **Kennesaw State University** — *January 2022 – May 2022, January 2024 – Present*
  Mechatronics Engineering, B.S. – Graduation December 2026 — CGPA: 4.00
- **Marine Corps Communication Electronics School** — *August 2022 – October 2023*
  Ground Radio Repair School — CGPA: 4.00
- **Chattahoochee Technical College** — *August 2020 – December 2021*
  General Education (Undecided Major) — CGPA: 3.80

Research & Engineering Experience

- **River Debris Trajectory Prediction using Machine Learning** — *June 2026 - August 2026*
  Funded Undergraduate Researcher – Summer Undergraduate Research Program, Dr. Matt Marshall
  - **Overview:** Independently designing a real-time vision system that detects, tracks, and forecasts the trajectories of floating river debris from a fixed riverbank camera, so a collection platform can be directed to where debris will be.
  - **Approach:** Measures surface velocity via dense optical-flow velocimetry, pairing it with object motion and slip in a CNN + LSTM-DNN forecaster whose Gaussian head yields calibrated open-loop multi-step forecasts over YOLO26 detections. Object manager reconciles short and long-term identity, reusing forecasts to reabsorb lost tracks.
  - **Publication:** First-author manuscript in preparation, targeting IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2027.
- **MEAP – Mass Estimation Autonomous Pathfinding for Search and Rescue** — *November 2025 - Present*
  Robotics Software Architect & Project Coordinator – AgriSys Lab, Dr. Muhammad Hassan Tanveer
  - **Overview:** Designing a UAV–UGV collaborative search-and-rescue system: the UAV surveys the environment while the UGV chooses advantageous traversable obstacles to clear with an onboard manipulator to shorten the path to each objective.
  - **Contribution:** Leading the UAV autonomy stack: flight motion framework, aerial map stitching, and the pipeline that turns survey passes into one searchable map. Also leading the ML pipeline estimating obstacle mass from sensor-fused SLAM point clouds to determine what the manipulator can safely move.
  - **Publication:** First-author manuscript in preparation, targeted for submission December 2026.
- **DARB – Drone Arm Robotic Backpack (Personal Project)** — *May 2025 - Present*
  Sole Designer and Builder
  - **Overview:** Designing and building an autonomous drone docking system: a robotic arm on a wearable backpack platform captures a drone mid-flight, stows it, and relaunches it.
  - **Approach:** Built a custom airframe that uses computer vision to align to IR LED markers on the platform, triggering arm capture once orientation converges.
  - **Hardware:** Custom airframe on a Matek H743 flight controller running ArduPilot, with encoder-driven NEMA 23 steppers actuating the capture arm.
- **SoilBus – Soil Monitoring Probe Network** — *August 2025 - December 2025*
  Lead Software Developer – AgriSys Lab, Dr. Muhammad Hassan Tanveer
  - **Overview:** Built a hub-less soil-monitoring mesh network for farmers: probes auto-discover neighbors and any single probe can offload the entire network's data.
  - **Results:** Used breadth-first search for inter-probe routing; validated across 5 physical nodes, holding to 26 simulated nodes before degradation.
  - **Publication:** SoilBus: An Ad Hoc Soil Sensor Network for Precision Agriculture.

Industry & Military Experience

- **EOSYS Industrial and Manufacturing System Integrator** — *May 2025 – August 2025*
  COOP — Marietta GA
  - **HMI tooling:** Built a Human Machine Interface display generator for P&IDs.
  - **Controls:** Designed and modified PLC programs.
- **United States Marine Corps Reserve** — *August 2022 - Present*
  Ground Radio Repair Specialist — Marietta GA
  - **Radio systems:** Troubleshoot, maintain, and operate USMC radio systems in time-sensitive conditions; active security clearance.
  - **Squad Leader:** 4th Combat Readiness Regiment; training, accountability, and readiness of a squad of Marines.

Technical Skills

Perception & ML: PyTorch, TensorFlow, YOLO, CNNs, LSTMs, OpenCV, ORB-SLAM3, sensor fusion, TensorRT
Autonomy & Robotics: ROS 2, Nav2, ArduPilot, PX4, Mission Planner, path planning, UAV/UGV integration
Embedded & Hardware: ESP32, Arduino, Raspberry Pi 5, Jetson Orin Nano, OpenMV, LiDAR, RGB-D cameras, IMUs, ad hoc networking
Manipulators & Automation: Denavit-Hartenberg parameters, forward and inverse kinematics, trajectory planning, Ladder Logic, TIA Portal, PLCs, HMI development
Languages & Tools: C++, Python, MATLAB, LaTeX, Linux/WSL2, Git, VS Code, SolidWorks, Onshape